AI Assisted Coding

Assignment - 3

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Question 1: Zero-Shot Prompting (Palindrome Number Program)

Write a zero-shot prompt (without providing any examples) to generate

a Python function that checks whether a given number is a palindrome

Code:

```python
# Generate python code for plaindrome verification in zero shot prompting
def is_palindrome(s):
    s = s.replace(" ", "").lower()  # Remove spaces and convert to lowercase
    return s == s[::-1]  # Check if the string is equal to its reverse
if __name__ == "__main__":
    user_input = input("Enter a string: ")
    if is_palindrome(user_input):
        print(f'"{user_input}" is a palindrome.')
    else:
        print(f'"{user_input}" is not a palindrome.')
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/AppData/
Desktop/All courses/AI assistant/lab2.py"
Enter a string: 12321
"12321" is a palindrome.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant>
```

Question 2: One-Shot Prompting (Factorial Calculation)

Write a one-shot prompt by providing one input-output example and

ask the AI to generate a Python function to compute the factorial of a

given number.

Code:

```python
# Generate a python program for factorial calculation using one shot prompting
#Example: Input: 5 Output: 120
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
if __name__ == "__main__":
    num = int(input("Enter a number to calculate its factorial: "))
    result = factorial(num)
    print(f"The factorial of {num} is {result}.")
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/A
Desktop/All courses/AI assistant/lab2.py"
Enter a number to calculate its factorial: 6
The factorial of 6 is 720.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant>
```

Question 3: Few-Shot Prompting (Armstrong Number Check)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python function to check whether a given number is an Armstrong number.

Code:

```python
# write a python program to generate Amstrong numbers using few shot prompting
#Example: Input: 153 Output: True
#Example: Input: 123 Output: False
#Example: Input: 370 Output: True
def is_armstrong(number):
    num_str = str(number)
    num_digits = len(num_str)
    sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)
    return sum_of_powers == number
if __name__ == "__main__":
    num = int(input("Enter a number to check if it's an Armstrong number: "))
    if is_armstrong(num):
        print(f"{num} is an Armstrong number.")
    else:
        print(f"{num} is not an Armstrong number.")
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/Ap
Desktop/All courses/AI assistant/lab2.py"
Enter a number to check if it's an Armstrong number: 153
153 is an Armstrong number.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/Ap
Desktop/All courses/AI assistant/lab2.py"
Enter a number to check if it's an Armstrong number: 123
123 is not an Armstrong number.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant>
```

Question 4: Zero-Shot Prompting (Perfect Number Check)

Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number.

Code:

```python
# Generate python Program to check perfect number using zero shot prompting
def is_perfect_number(n):
    sum_of_divisors = sum(i for i in range(1, n) if n % i == 0)
    return sum_of_divisors == n
if __name__ == "__main__":
    number = int(input("Enter a number to check if it's a perfect number: "))
    if is_perfect_number(number):
        print(f"{number} is a perfect number.")
    else:
        print(f"{number} is not a perfect number.")
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/Ap
Desktop/All courses/AI assistant/lab2.py"
Enter a number to check if it's a perfect number: 12
12 is not a perfect number.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/Ap
Desktop/All courses/AI assistant/lab2.py"
Enter a number to check if it's a perfect number: 6
6 is a perfect number.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant>
```

Question 5: Few-Shot Prompting (Even or Odd Classification with Validation)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python program that determines whether a given number is even or odd, including proper input validation.

Code:

```python
# Generate python program to check Even or Odd Classification with Validation using few shot prompting
#Example: Input: 4 Output: Even
#Example: Input: 7 Output: Odd
#Example: Input: -2 Output: Even
def check_even_odd(num):
    if not isinstance(num, int):
        return "Invalid input. Please enter an integer."
    return "Even" if num % 2 == 0 else "Odd"
if __name__ == "__main__":
    try:
        user_input = int(input("Enter an integer to check if it's Even or Odd: "))
        result = check_even_odd(user_input)
        print(f"{user_input} is {result}.")
    except ValueError:
        print("Invalid input. Please enter a valid integer.")
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/AppData/Local/Programs/Py
Desktop/All courses/AI assistant/lab2.py"
Enter an integer to check if it's Even or Odd: 12
12 is Even.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/AppData/Local/Programs/Py
Desktop/All courses/AI assistant/lab2.py"
Enter an integer to check if it's Even or Odd: 77
77 is Odd.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant> & "C:/Users/NITHIN REDDY/AppData/Local/Programs/Py
Desktop/All courses/AI assistant/lab2.py"
Enter an integer to check if it's Even or Odd: -55
-55 is Odd.
PS C:\Users\NITHIN REDDY\OneDrive\Desktop\All courses\AI assistant>
```